

Last Time:

- Col vec of size n = n -tuple of $\mathbb{R}$ #'s

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = (a_1, \dots, a_n) \neq [a_1, \dots, a_n] \text{ row vec}$$

(has commas) (no commas)

- Defined vector space ^{v.s.} via (VS1) – (VS8) axioms.
- Exs of v.s.: $\mathbb{R}^n$, $\mathcal{F}(X, \mathbb{R}) = \{f: f \text{ fcn } X \rightarrow \mathbb{R}\}$,

$$M_{m \times n}(\mathbb{R}) = \left\{ \underbrace{\begin{bmatrix} A_{11} & \dots & A_{1n} \\ \vdots & & \vdots \\ A_{m1} & \dots & A_{mn} \end{bmatrix}}_{m \times n \text{ matrix}} : A_{ij} \in \mathbb{R} \quad \forall i \text{ \& } \forall j \right\}$$

- Cancellation Law: V v.s., $x, y, z \in V$, $x + z = y + z$
 $\Rightarrow x = y$.

Today: Continue §1.2 Vector Spaces and Subspaces

Cor 1.16: V v.s.

(1) V has a unique ^{0_V} zero vec.

(2) $\forall v \in V$, v has unique ^{inv} additive inverse. Call it $-v$.

Pf: (1): See Unit 1.2 Notes.

End of Part
of Pf 

(2): Suppose w_1, w_2 are additive inverses of v .

Want ^{WTS} to show: $w_1 = w_2$.

$$\underbrace{v + w_1}_{VS4} = \underbrace{0_V}_{VS4} = \underbrace{v + w_2}_{VS1} \Rightarrow \underbrace{w_1 + \underbrace{v}_{\text{blue}}}_{\text{red}} = \underbrace{w_2 + \underbrace{v}_{\text{blue}}}_{\text{purple}}, \text{ so}$$

Cancellation Law $\Rightarrow w_1 = w_2$.

End of Pf
 $\square$

Prop 1.17: V v.s. Then $\forall v \in V$ and $\forall a \in \mathbb{R}$,

$$(1) 0 \cdot v = 0_V.$$

$$(2) (-a) \cdot v = -(av) = a(-v) \quad (\text{so } a = -1 \Rightarrow -v = (-1) \cdot v) \quad \text{VS5}$$

$$(3) a \cdot 0_V = 0_V$$

Pf: (1) & (2): See Pf of Thm 1.2 in textbook. ^{FIS}

$$(3): \underbrace{a \cdot 0_V}_{\text{purple}} + \underbrace{0_V}_{\text{blue}} \stackrel{VS3}{=} \underbrace{a \cdot 0_V}_{\text{purple}} \stackrel{VS3}{=} a \cdot (\underbrace{0_V}_{\text{red}} + \underbrace{0_V}_{\text{blue}}) \stackrel{VS7}{=} \underbrace{a \cdot 0_V}_{\text{blue}} + \underbrace{a \cdot 0_V}_{\text{purple}},$$

so Cancellation Law $\Rightarrow 0_V = a \cdot 0_V$. $\square$

Rmk: (1): intuitions about 0-vecs and additive inverses from $\mathbb{R}^n$ are true for any vector space!

(2): From now on, $0 \in V$ means $0_{\underbrace{V}_{\text{red}}}$.
unless V needs to be emphasized

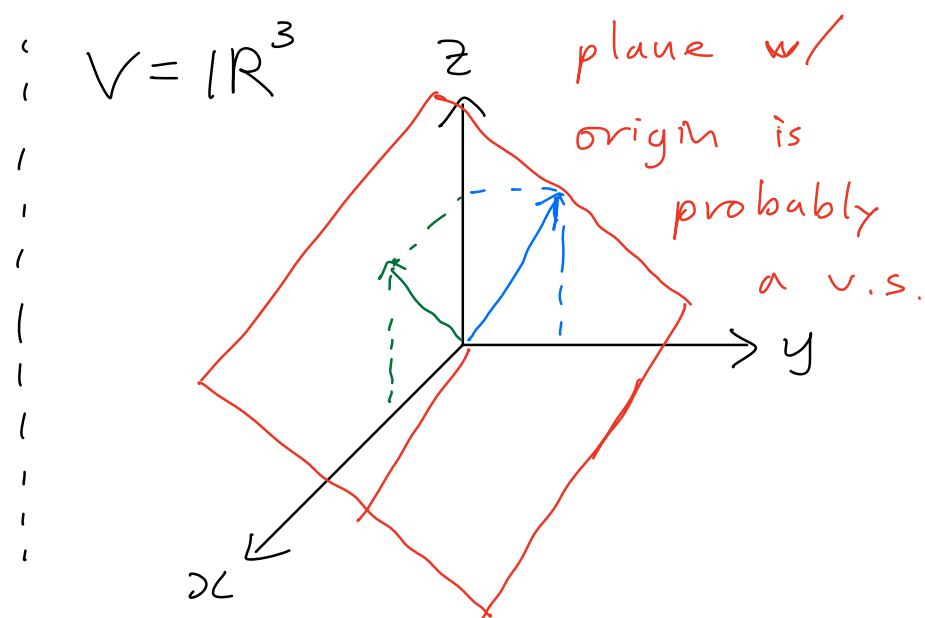
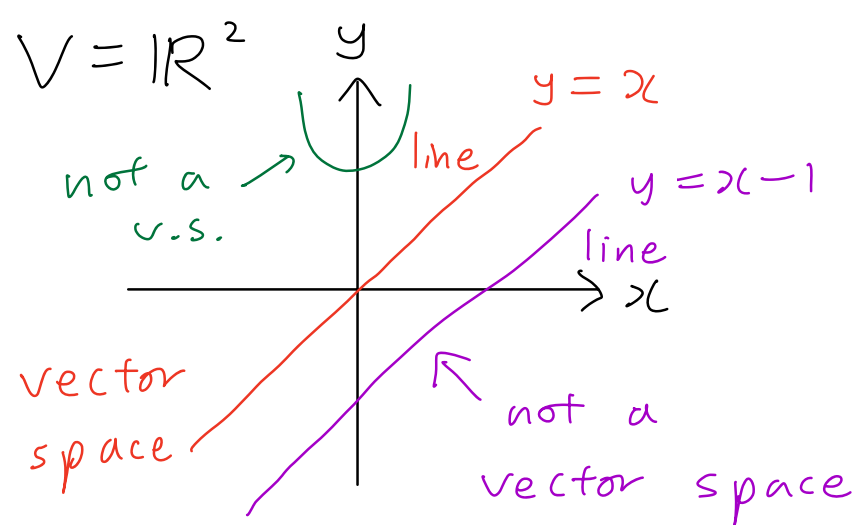
Subspaces

Motivation: How do we go beyond $\mathbb{R}^n$, $\mathcal{F}(X, \mathbb{R})$,

$M_{m \times n}(\mathbb{R})$ to find more Exs of v.s.? Subsets?



When is a subset W of a given v.s., equipped w/ the same $+$ & $\cdot$ operations, also a v.s.?



Def 1.18: V v.s. Then $W \subseteq V$ is a subspace of V if W w/ induced $+$ & $\cdot$ from V is a v.s.

I.e., given $+: V \times V \rightarrow V$ and $\cdot: \mathbb{R} \times V \rightarrow V$, then W w/ restricted operations

$+_W: W \times W \rightarrow W$ and $\cdot_W: \mathbb{R} \times W \rightarrow W$ satisfies

$w_1, w_2 \mapsto w_1 + w_2$ $c, w \mapsto c \cdot w$

(VS1) - (VS8).

Ex 1.19:

(1) $V, \{0\} \subseteq V$ are subspaces. $\{0\}$ is called the trivial subspace.

(2) In $\mathbb{R}^3$, consider subset $W := \{(x, y, 0) : x, y \in \mathbb{R}\}$, xy -plane

Then $\exists$ ^{1-1 & onto fcn} bijection $\mathbb{R}^2 \longleftrightarrow W$ where
 $(x, y) \mapsto (x, y, 0)$

$+_W$ corresponds to $+\mathbb{R}^2$ and $\cdot_W$ corresponds to $\cdot\mathbb{R}^2$.

B/c $\mathbb{R}^2$ is a v.s., so is W w/ $+_W$ & $\cdot_W$

$\Rightarrow W$ is a subsp of $\mathbb{R}^3$.

Before giving more Exs of subsp, need simpler way to check when a given $W \subseteq V$ is a subsp:

Thm 1.20 (Subsp ^{Crit} Criterion): V v.s., $W \subseteq V$. Then

W is a subsp of V if and ^{iff} only if:

(1) $0_V \in W$

(2) $x + y \in W \quad \forall x, y \in W$ (where $+: V \times V \rightarrow V$)

(3) $c \cdot x \in W \quad \forall c \in \mathbb{R} \text{ and } \forall x \in W$

(where $\cdot: \mathbb{R} \times V \rightarrow V$)

Pf: FIS Thm 1.3. Idea: $(\Rightarrow)$ direction is

just the def, ^{& Prop 1.17(3)} and $(\Leftarrow)$ direction verifies that

$+_W$ & $\cdot_W$ (that's equal to $+$ & $\cdot$ on correct domain)

satisfies (VS1) – (VS8) by using $+$ & $\cdot$. $\square$

Actually more important to learn how to use statements than the Pfs!

Rmk 1.21: Pf of Subsp Crit implies that:

(1) $0_W = 0_V$

(2) additive inverse of $x \in W$ wrt with respect to W is $-x = (-1) \cdot x$ by Prop 1.17, which coincides w/ additive inverse of x wrt V .

Ex 1.22: Consider $\mathbb{R}^n$ and

$W := \{(a_1, \dots, a_n) \in \mathbb{R}^n : \sum_{i=1}^n a_i = 0\}$. Then W is a subsp (of $\mathbb{R}^n$).

Pf: Need to check 3 conditions from Subsp Crit.

(1): $0_{\mathbb{R}^n} = (0, \dots, 0)$ satisfies $\sum_{i=1}^n 0 = 0$ (sum of coords is 0)

$\Rightarrow 0_{\mathbb{R}^n} \in W$ by def of W . ◻

indices can be skipped if $\nexists$ confusion
(2): $\forall (a_1, \dots, a_n), (b_1, \dots, b_n) \in W$, $\sum_{i=1}^n a_i = 0 = \sum_{i=1}^n b_i$

by def of W . So $\sum (a_i + b_i) = \sum a_i + \sum b_i = 0 + 0 = 0$

$$\Rightarrow (a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n) \in W. \quad \square$$

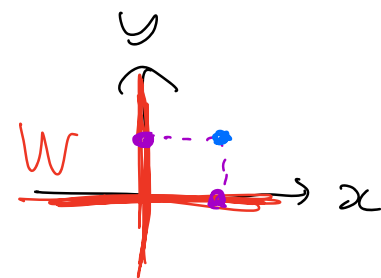
$$(3): \forall c \in \mathbb{R} \text{ and } \forall (a_1, \dots, a_n) \in W, \quad 0 = \sum a_i$$

$$\Rightarrow \sum (ca_i) = c \sum a_i = c \cdot 0 = 0, \quad \text{so}$$

$$c(a_1, \dots, a_n) = (ca_1, \dots, ca_n) \in W. \quad \square$$

Ex 1.23: In $\mathbb{R}^2$, consider subset

$$W := \{(a, b) \in \mathbb{R}^2 : a=0 \text{ or } b=0\}$$



Then W is not a subsp, because ^{b/c} $(1, 0), (0, 1) \in W$

but $(1, 0) + (0, 1) = (1, 1) \notin W$ by def, violating

Subsp Crit (2).